Modeling a Multi-Band Equalizer for Audio Enhancement and Spectral Analysis

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***Abstract*—This project presents the design and implementation of a multi-band audio equalizer using continuous-time band-pass filters modeled with RLC circuits. The equalizer decomposes an input signal into five frequency-selective components, applies independent gain adjustments to each band, and recombines the filtered signals to produce a modified output. The frequency response and impulse response of each filter stage were analyzed to verify the intended band-pass behavior. Gain presets were then used to demonstrate common equalizer configurations, including bass boost, treble boost, and a near-unity response. The system was applied to both music recordings and an environmental audio recording containing bird vocalizations. Spectrogram analysis shows that the equalizer enhances the visibility of bird chirps by amplifying frequency bands associated with vocalizations while attenuating background noise. These results demonstrate that a parallel filter-bank architecture provides an effective method for frequency-selective audio enhancement and acoustic signal analysis.**

I. Background

Audio recordings often contain unwanted noise that can degrade the quality of the sound. The ability to selectively amplify or attenuate specific frequency bands is fundamental to modern audio processing systems, including equalizers found in smartphones, music players, and professional audio equipment. By modifying the spectral content of a signal, equalization can enhance clarity, tonal balance, and suppress unwanted noise.

A multi-band equalizer is a powerful tool for enhancing audio by allowing users to adjust the amplitude of specific frequency bands. This is accomplished by decomposing an input signal into several frequency-selective components using filters such as low-pass, high-pass, band-pass, and band-stop filters. Each band is independently scaled by a gain factor before recombining the filtered signals. This structure enables targeted control over bass, midrange, and treble content while preserving overall signal integrity.

Frequency-selective filtering can be implemented using continuous-time circuit models, which provide well-characterized impulse and frequency responses. First-order RC filters allow straightforward analytical modeling of magnitude and phase behavior, making them suitable for constructing modular equalizer stages. When multiple filters operate in parallel and are combined through weighted summation, the resulting system achieves customizable spectral shaping.

Equalization additionally plays a critical role in environmental audio analysis. Natural recordings, such as bird vocalizations, often include wind noise and broadband interference that obscure important biologically relevant features. Spectrogram analysis, the visualization of time-varying frequency content, enables identification of recurring vocalization patterns. Through appropriate frequency bands and gain settings, an equalizer can isolate and amplify bird calls while reducing background noise, facilitating more accurate species identification and behavioral studies.

This project focuses on modeling and designing a continuous-time multi-band equalizer to enhance music recordings and improve the clarity and identification of bird vocalizations in noisy environments through spectral analysis.

II. Methods

A. Formulation

Let $h_i(t)$ be the impulse response of an linear time-invariant (LTI) system that is designed to pick out a specific band of frequencies. Figure 1 shows five of these systems that pick out a band of frequency. Note that the frequency ranges are obtained as a result of parameters discussed later in the design.

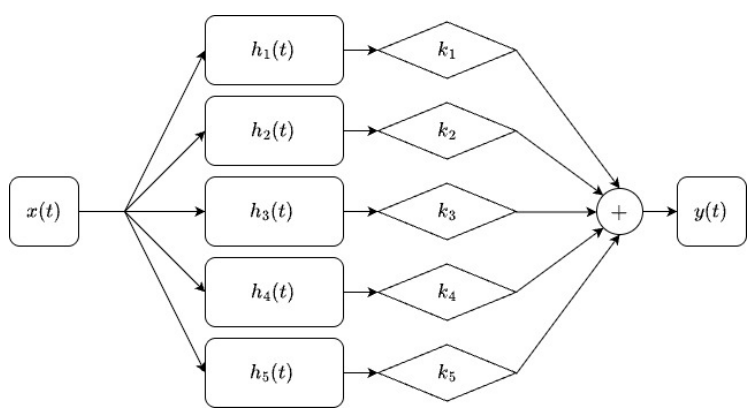

Fig. 1. Equalizer system design.

In order to implement $h_i(t)$, we use a band-pass filter. Resistor-inductor-capacitor (RLC) circuits are networks that have long been used as a method to implement band-pass filters. We can derive transfer functions using the impedance

of the devices, and because these devices are linear, they can be applied to each sinusoidal component of an input function. Consider the following circuit:

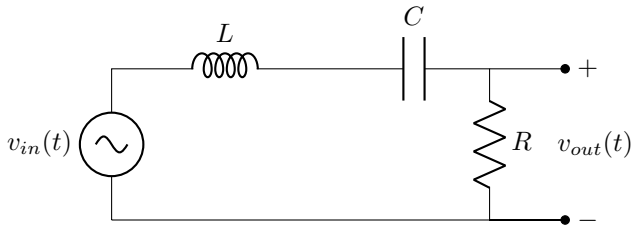

Fig. 2. Series RLC band-pass filter with output measured across the resistor.

This circuit picks out one frequency band and attenuates others based on the values passed into R, L, and C. Equation 1 is the i'th's circuit's transfer function where $i = 1, 2, ..., 5$ (see Appendix II.A for derivation).

$$X_i(jw) = \frac{jw\frac{R}{L}}{(j\omega)^2 + j\omega\frac{R}{L} + \frac{1}{LC}} \tag{1}$$

Looking at Equation 1, we can see that if $\omega = 0$, $X_i(j\omega) = 0$, and if $\omega \to \infty$, $X_i(j\omega) \to 0$. Thus we know that this filter is neither lowpass nor highpass.

Next, we will introduce the continuous time linear constant coefficient differential equation (LCCDE) for this circuit, which is given by (see Appendix II.B for derivation).

$$\frac{dv_{in}}{dt} = \frac{L}{R}\frac{d^2v_{out}}{dt^2} + \frac{dv_{out}}{dt} + \frac{1}{RC}v_{out} \tag{2}$$

Different circuit parameters were chosen for each task. This is because in each task, we want to pick different bands, and within those bands, sometimes we may need to limit the bandwidth. We controlled the center frequency by using the fact that $\omega_c = \frac{1}{\sqrt{LC}}$. In order to control the bandwidth, we utilized the idea of the Q-factor of the circuit, given by

$$Q = \frac{w_c L}{R} \tag{3}$$

Intuitively, the Q factor measures how well the circuit can "pick" out its frequency range. For a higher Q-factor, the bandwidth is smaller, and vice versa. Because $Q \sim R^{-1}$, we can increase the bandwidth by increasing R, and vice versa.

The optimal input parameters R, L, and C for each circuit and each task were determined qualitatively through trial and error and are listed in Appendix II.C.

B. Song Task

In this task, we aim to minimize loss of frequency bands while also separating bands into distinct categories. Figure 3 lists the original system with the frequency ranges added on. Note that the ranges are categorized by their half-power cutoff frequencies.

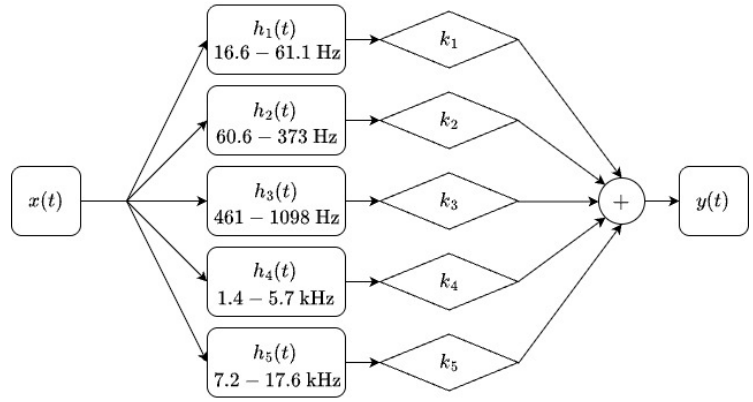

Fig. 3. Equalizer system design with song task frequency ranges

$\vec{k} = (k_1, ..., k_5)$ is a gain vector that is customization within our code. For each modification within the song task, we list $\vec{k}$ in Appendix II.D.

C. Bird Task

For the bird task, we want narrower bands around precise frequencies. We found these bans by looking into at the spectrogram of the bird audio and finding where the bird chirps are (see Figure x for a zoomed in view).3

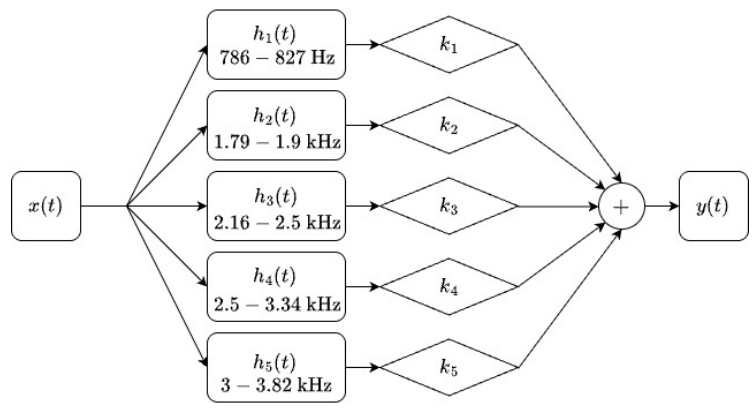

Fig. 4. Equalizer system design with bird task frequency ranges

III. Results

The performance of the designed equalizer was evaluated by examining the frequency response of the individual filters, the combined system response for different preset gain configurations, and the enhancement of bird vocalizations in a noisy recording.

A. Individual Filter Responses

Figure **??** shows the magnitude and phase responses of the five band-pass filters used in the equalizer design. Each filter isolates a specific portion of the frequency spectrum corresponding to the bands identified during system design. The magnitude plots show clear peaks at the intended center frequencies, confirming that each RLC circuit successfully

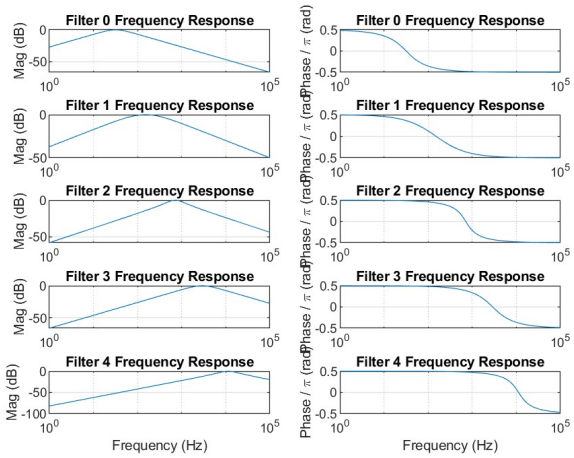

Fig. 5. Magnitude and phase responses of the five band-pass filters used in the equalizer. Each filter isolates a specific frequency band.

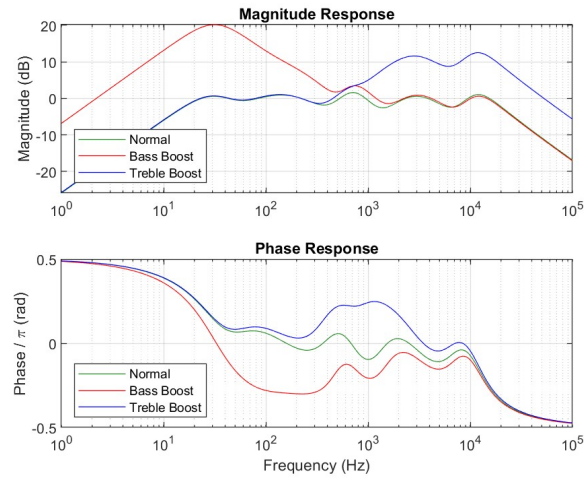

Fig. 6. Overall equalizer magnitude and phase responses for three preset gain configurations: normal, bass boost, and treble boost.

isolates its target band while attenuating frequencies outside the passband.

The phase responses demonstrate the expected phase transitions typical of band-pass filters. Together, these results confirm that the individual filter stages operate as intended and provide the frequency selectivity necessary for the equalizer architecture.

B. Equalizer Preset Responses

To evaluate the behavior of the overall system, three preset gain configurations were implemented: normal response, bass boost, and treble boost. Figure **??** shows the resulting magnitude and phase responses of the full equalizer system.

The bass boost configuration increases gain in the lower frequency bands while attenuating higher frequencies, producing a stronger response in the low-frequency region. Conversely, the treble boost configuration increases gain in higher frequency bands while reducing lower-frequency components. The normal preset maintains a relatively flat frequency response across the audible range. These results demonstrate that the gain vector effectively controls the spectral shaping of the signal.

C. Bird Vocalization Enhancement

The equalizer was applied to a field recording containing bird vocalizations embedded in environmental noise. Figure **??** shows spectrograms of the original recording and the filtered signal after applying the designed equalizer. In the original spectrogram, broadband noise is present across much of the lower frequency spectrum, which reduces the visibility of short-duration vocalization events.

The boxed region highlights a time interval where multiple bird chirps occur within a narrow frequency band. In the original recording, these chirps appear faint and partially masked by background noise. After equalization, the same region shows significantly higher intensity and clearer vertical structures

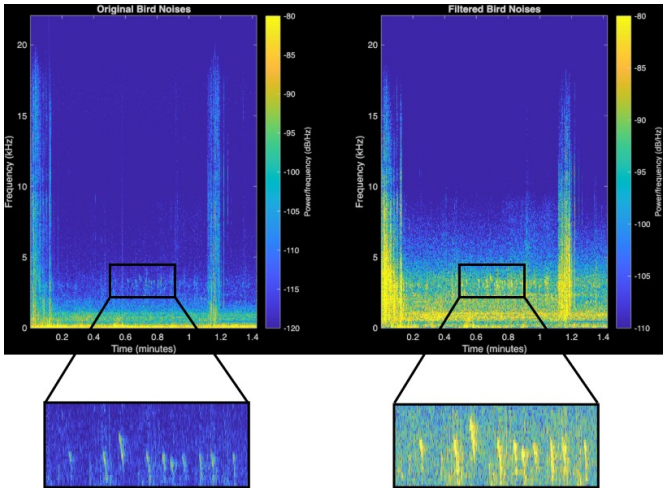

Fig. 7. Spectrogram comparison of the original bird recording (left) and the filtered signal (right). The zoomed region highlights improved visibility of bird vocalizations after equalization.

corresponding to individual chirps. The zoomed-in panels emphasize that the filtered signal exhibits stronger spectral contrast between the bird vocalizations and the surrounding noise floor.

This improvement occurs because the equalizer selectively amplifies the frequency bands where bird calls were observed while attenuating lower-frequency noise components. As a result, individual vocalization events become easier to distinguish in both time and frequency. The enhanced visibility of these chirps allows recurring vocalization patterns to be more readily identified and analyzed within the spectrogram.

Overall, the results demonstrate that the multi-band equalizer successfully isolates frequency bands, allows flexible spectral shaping through gain presets, and enhances biologically relevant acoustic features in environmental recordings.

IV. Conclusion

In this project, a multi-band equalizer was designed and implemented using continuous-time band-pass filters modeled with RLC circuits. The system decomposes an input signal into several frequency-selective components, applies independent gain adjustments to each band, and recombines the filtered signals to produce a modified output. By carefully selecting the circuit parameters and gain vectors, the equalizer was able to shape the spectral content of the signal and emphasize specific frequency ranges.

Analysis of the individual filter responses confirmed that each filter successfully isolated its intended frequency band. The overall system frequency response demonstrated that different gain presets could effectively modify the spectral balance of the signal, producing bass-boost and treble-boost behaviors consistent with typical audio equalizers.

Application of the equalizer to the bird vocalization recording demonstrated the practical effectiveness of the design. Spectrogram analysis showed that the filtered signal improved the visibility of bird chirps by reducing background noise and increasing the prominence of frequency bands associated with vocalizations. In particular, the enhanced contrast in the boxed spectrogram region highlighted recurring chirp structures that were less distinguishable in the original recording.

Overall, the results demonstrate that a parallel filter-bank architecture combined with adjustable gain coefficients provides a flexible and effective approach for audio signal enhancement. This project illustrates how classical circuit-based filtering concepts can be applied to digital signal processing tasks such as noise reduction and acoustic feature isolation.

References

[1] A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1997.

[2] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2010.

[3] S. W. Smith, *The Scientist and Engineer's Guide to Digital Signal Processing*. San Diego, CA, USA: California Technical Publishing, 1997.

[4] L. R. Rabiner and B. Gold, *Theory and Application of Digital Signal Processing*. Englewood Cliffs, NJ, USA: Prentice Hall, 1975.

Appendix

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The following figures provide additional characterization of the individual filter stages and the overall equalizer system. These plots supplement the results discussed in the main body of the report and illustrate the behavior of the designed band-pass filters and the resulting system response.

A. Impulse Responses of Individual Filters

Figure **??** shows the impulse responses of the five band-pass filters used in the equalizer architecture. Each impulse response corresponds to the time-domain behavior of the associated RLC circuit described in Section II. The responses exhibit damped oscillatory behavior consistent with second-order band-pass systems, with decay rates determined by the circuit parameters and Q-factor.

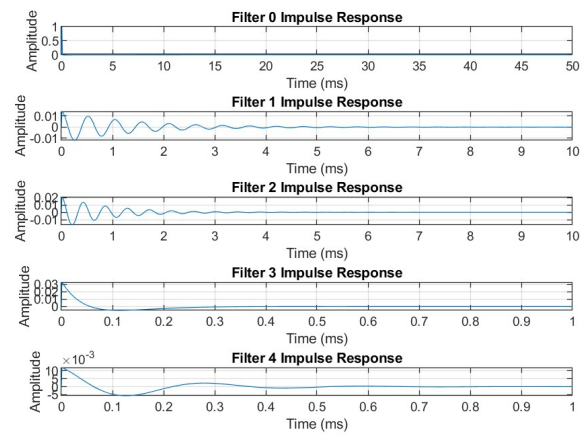

Fig. 8. Impulse responses of the five band-pass filters used in the equalizer design.

B. Additional Filter Frequency Responses

Figure **??** shows additional magnitude and phase responses of the individual filters. These plots further confirm the band-pass behavior of each filter stage and illustrate the separation of frequency bands across the system.

C. Overall Equalizer Response

Figure **??** shows the overall magnitude and phase response of the equalizer when all filter stages and gain weights are combined. The response reflects the cumulative spectral shaping produced by the parallel filter architecture and gain vector applied in the system.

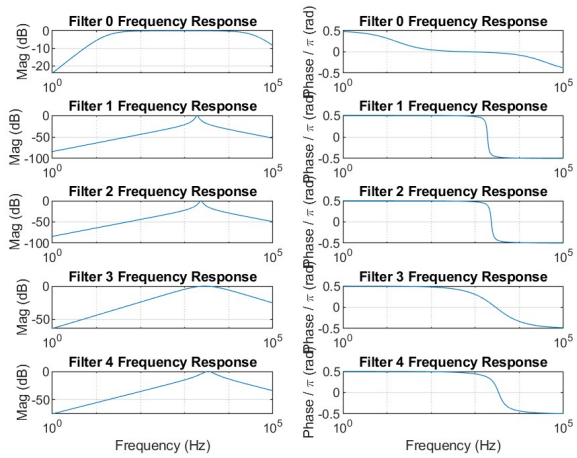

Fig. 9. Additional magnitude and phase responses for the individual band-pass filters.

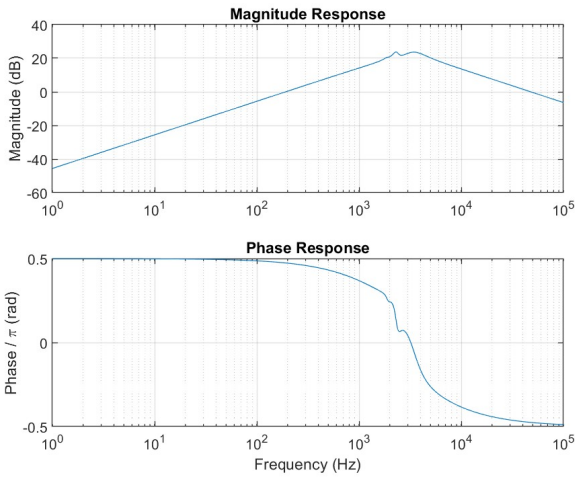

Fig. 10. Overall equalizer magnitude and phase response for the combined filter system.

APPENDIX A
II. METHODS

A. Circuit Frequency Response Derivation

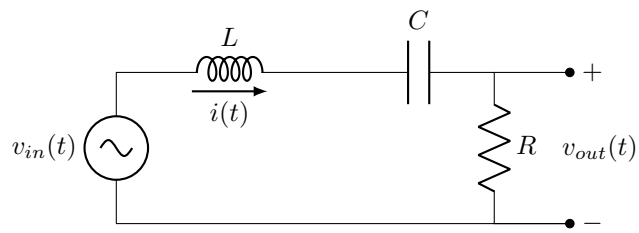

Fig. 11. Time Domain Series RLC band-pass filter with output measured across the resistor.

First, we want to convert this circuit to the frequency domain using the impedance of the devices in the circuit.

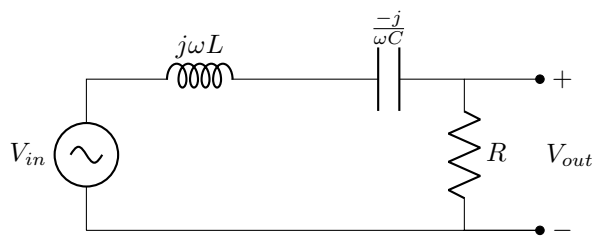

Fig. 12. Frequency domain equivalent.

Now, we can perform voltage division across the resistor in order to obtain V_{out} in terms of V_{in}.

$$V_{out} = V_{in}\frac{R}{R + j\omega L - \frac{j}{\omega C}} \tag{4}$$

Multiply everything by $\frac{j\omega}{L}$.

$$V_{out} = V_{in}\frac{jw\frac{R}{L}}{(j\omega)^2 + j\omega\frac{R}{L} + \frac{1}{LC}} \tag{5}$$

We know that the frequency response is the ratio of the output and input voltages. So,

$$X(jw) = \frac{V_{out}}{V_{in}} = \frac{jw\frac{R}{L}}{(j\omega)^2 + j\omega\frac{R}{L} + \frac{1}{LC}} \tag{6}$$

B. Circuit Continuous Time LCCDE Derivation

From the circuit in Figure x, perform Kirchhoff's Voltage Law.

$$v_{in}(t) = L\frac{di}{dt} + \frac{1}{C}\int i(t)dt + iR \tag{7}$$

Now, differentiate both sides.

$$\frac{dv_{in}}{dt} = L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{1}{C}i \tag{8}$$

We know that $v_{out} = iR \implies i = \frac{v_{out}}{R}$. So,

$$\frac{dv_{in}}{dt} = \frac{L}{R}\frac{d^2v_{out}}{dt^2} + \frac{dv_{out}}{dt} + \frac{1}{RC}v_{out} \tag{9}$$

C. Optimal Circuit Parameters

UPDATE THESE

i	R (Ω)	L (mH)	C (μF)
1	140	500	50
2	220	112	10
3	200	39	1
4	800	30	0.1
5	1300	20	0.01

TABLE I
OPTIMAL CIRCUIT PARAMETERS FOR SONG TASK

i	R (Ω)	L (mH)	C (μF)
1	33.3	500	50
2	26.5	112	10
3	39.5	39	1
4	50	30	0.1
5	50	20	0.01

TABLE II
OPTIMAL CIRCUIT PARAMETERS FOR BIRD TASK

D. Gain Vectors

- Normal: $\vec{k} = (0.8, 1, 0.9, 0.95, 0.9)$
- Bass boost: $\vec{k} = (10, 1, 1, 0.95, 0.82)$
- Treble boost: $\vec{k} = (0.8, 1, 0.5, 3.5, 3.5)$